Calibration Degradation in Fused ECFP Pipelines and Temperature Scaling Mitigation

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Technical Report for Cold-Drug DDI Benchmark

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Abstract

Calibration degradation after fusing Extended-Connectivity Fingerprint (ECFP) features is a predictable side-effect of increased effective model capacity, heterogeneous feature sources that create conflicting evidence and region-dependent confidence errors, and distributional shifts that the fusion model amplifies rather than averages out. This report provides a comprehensive analysis of why calibration worsens when ECFP4 fingerprints are fused with graph attention network embeddings in the context of cold-drug DDI prediction, and presents a systematic mitigation strategy centered on post-hoc temperature scaling. We define key concepts, identify six mechanistic causes of calibration failure, survey the relevant evidence base, recommend diagnostic metrics and chart types, and provide an implementation blueprint for temperature scaling on fused ECFP outputs. The report concludes with practical recommendations for writing a calibration-aware evaluation chapter.

Keywords: calibration, ECFP, temperature scaling, drug–drug interaction, multimodal fusion, reliability diagram

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1 Executive Summary

A tight, defensible mitigation strategy involves three stages. *First*, diagnose *how* calibration breaks: global overconfidence vs. classwise vs. subsetwise; i.i.d. vs. shifted. Use reliability diagrams plus scalar metrics (ECE, MCE, Brier, NLL) and report binning choices and uncertainty around estimates. *Second*, apply post-hoc temperature scaling to the final fused logits first; it is low-variance (1 parameter), preserves accuracy and argmax in multiclass settings, and is a standard baseline in modern calibration work. *Third*, if miscalibration is not mostly global—e.g., it varies by class, interaction type, cold-start subset, or confidence level—use structured extensions: classwise or vector scaling, adaptive or local temperature scaling, or (when calibration data suffice) isotonic or Dirichlet-style calibrators. Calibration must be validated under the fusion regimes of interest (e.g., cold-drug splits, temporally shifted splits, scaffold splits), because calibration that looks good i.i.d. can fail under shift; ensembles are often more robust than post-hoc scaling under shift, but at higher compute cost.

2 Definitions and Scope

2.1 Extended-Connectivity Fingerprints (ECFP)

ECFP are circular, neighbourhood-expansion fingerprints designed for structure–activity modelling. They iterate over atom environments up to a specified radius, hash identifiers for local substructures, and fold them into fixed-length sparse bit or count vectors (e.g., 1024/2048/4096 bits). A key implementation detail for the fusion–calibration relationship: **folding increases hash collisions**, i.e., distinct substructures mapping to the same bit position, introducing structured noise and sometimes spurious correlations that a high-capacity fusion model can exploit with high confidence.

2.2 Unbindable Data Types

In this report, **unbindable data types** refers to two feature sources that cannot be meaningfully bound by a fixed, semantics-preserving alignment without learning a mapping or choosing an arbitrary binding operator. This includes two common chemical ML cases:

- **Pairwise entity fusion:** two molecules (drug A, drug B) each encoded as ECFP vectors, where the pair is unordered but representing it requires an ordering or a permutation-invariant operator.
- **Heterogeneous fusion:** ECFP (hashed sparse bits/counts) fused with learned graph embeddings, sequence embeddings, or knowledge-graph features, where feature scales, sparsity, and semantics differ.

This aligns with multimodal ML's core challenge: heterogeneity across modalities implies non-trivial alignment and fusion choices; naive fusion can create brittle decision rules and confidence behaviour.

2.3 Calibration

A probabilistic classifier is **calibrated** if predicted probabilities match empirical correctness frequencies (e.g., among predictions with confidence 0.8, approximately 80% are correct). Reliability diagrams visualise deviations; proper scoring rules (log loss, Brier) quantify probabilistic accuracy. In modern deep models, high accuracy does not imply calibration; architecture and training choices can systematically produce overconfident probabilities.

2.4 Temperature Scaling

Temperature scaling is a post-hoc method that rescales logits $\mathbf{z}$ by a single positive scalar T:

$$p_T(y{=}c \mid x) = \mathrm{softmax}\left(\frac{z_c(x)}{T}\right). \tag{1}$$

T is fitted on a held-out calibration set by minimising negative log-likelihood (NLL). In multiclass classification, it preserves argmax predictions (hence usually preserves accuracy) while adjusting confidence. It is a single-parameter special case of scaling approaches related to Platt-style calibration.

3 Mechanisms of Calibration Degradation

Calibration degradation in fusion is rarely mysterious; it typically falls out of the interaction between capacity, evidence aggregation, and mismatch/noise.

3.1 Statistical and Optimisation Causes

Cross-entropy training with hard labels pushes models to create large logit margins, yielding very confident softmax outputs even when the underlying problem contains ambiguity (aleatoric uncertainty) or the model is misspecified. Increasing effective capacity (depth/width) and certain training components (e.g., BatchNorm) have been empirically associated with worse calibration despite strong accuracy. Fusion often increases effective capacity (more inputs, more parameters, richer interactions), so it can improve discrimination while producing *sharper* probabilities—often overconfident unless regularised or calibrated.

3.2 Representational Causes Specific to ECFP

ECFP's hashed/folded representation introduces two properties that matter for calibration when fused:

- **Collision-induced ambiguity:** folding raises collision probability, causing multiple distinct substructures to share bits, which is a form of structured feature noise.
- **Sparsity and heavy-tailed feature counts:** bit vectors create sparse, high-variance activation patterns. When a fusion model learns that certain sparse patterns correlate with labels, it can become extremely confident on those patterns, even if the correlation is partly artefactual.

The net effect is that adding ECFP to another representation can increase the *perceived separability* of the training data, which increases confidence more than it increases true conditional correctness.

3.3 Feature Overlap and Evidence Double Counting

When two sources encode overlapping information (e.g., graph features and ECFP both derived from 2D structure), the model may **double count correlated evidence**, inflating confidence. Conversely, when sources conflict (e.g., noisy assay features vs. structure), naive fusion can still output high confidence because the model learns spurious rules for resolving conflicts. Fusion increases representation richness but also increases the space of shortcuts that yield large margins.

3.4 Distributional Shift and Split-Induced Mismatch

Calibration is distribution-dependent. Even if you temperature-scale on an i.i.d. validation split, calibration can deteriorate under chemical novelty (scaffold/cold-start splits), temporal drift (newer chemistry), or domain shift between datasets. This is particularly salient in DDI settings where new drugs differ from known drugs. Large-scale evaluations of predictive uncertainty under dataset shift show that post-hoc calibration can fall short under shift, and methods that marginalise over models (e.g., deep ensembles) are often more robust.

3.5 Label Noise and Incomplete Labels

Cheminformatics and pharmacology labels commonly contain measurement noise, extraction/curation noise, or incompleteness (unknown interactions treated as negatives). Hard-label cross-entropy can translate such noise into overconfident predictions; empirical work on mixup/label smoothing argues that overconfidence is partly a consequence of hard targets.

4 Summary of Causes, Signatures, and Mitigations

Table 1 summarises the six main causes of calibration degradation, their diagnostic signatures, and recommended mitigations ordered from simplest to most complex.

5 Evidence Base from Literature

5.1 Calibration in Neural Networks

Large empirical studies establish that many modern neural networks are miscalibrated and that temperature scaling is a strong, simple post-hoc baseline. Guo et al. demonstrated that as neural network depth and width increase, calibration degrades despite accuracy improvements, and that a single learned temperature parameter can substantially reduce ECE. Subsequent work revisiting calibration continues to treat temperature scaling as a key baseline across model families.

Table 1: Causes of calibration degradation in fused ECFP pipelines.

Cause	Calibration Signature	Diagnostic Test	Mitigation (simplest first)
Increased capacity via fusion	Global overconfidence; reliability curve below diagonal	Compare max-softmax histograms; NLL vs accuracy gap	Temperature scaling; stronger regularisation; label smoothing/mixup; ensembles
ECFP folding / collisions	Overconfident spikes on certain sparse patterns	Compare calibration by fingerprint density / collision proxy; ablate fpSize	Increase fpSize; count fingerprints; reduce folding; post-hoc calibration
Feature overlap (double counting)	Very high confidence with modest accuracy gains	Calibrate separately per modality branch; measure modality agreement	Late fusion + (re)calibration; gating with regularisation; ensemble fusion
Modality conflict / unbindable fusion	Region-dependent miscalibration; classwise errors	Conditional ECE by subset (cold-start vs warm, scaffold bins)	Classwise/vector scaling; local/adaptive TS; isotonic/Dirichlet calibration
Distribution shift	Calibration good i.i.d., bad OOD	Shifted test sets (temporal/scaffold/distribution change)	Ensembles; recalibration per domain; conformal prediction as backup
Label noise / incompleteness	Persistent miscalibration even after TS	Inspect label reliability; compare soft-label training	Label smoothing/mixup; Bayesian/noise-aware modelling; careful negatives

5.2 Multimodal and Fusion-Specific Evidence

Fusion can worsen trustworthiness properties even when it improves conventional performance. Dedicated multimodal calibration work formalises desiderata such as "confidence should not increase when one modality is removed" and evaluates calibration failures in multimodal settings. Recent clinical multimodal analyses similarly find that multimodal fusion can degrade selective prediction reliability.

5.3 Cheminformatics: Uncertainty and Calibration

In drug discovery contexts, multiple studies emphasise that uncertainty quality (including calibration) matters because experimental follow-up is expensive and models can be overconfident on out-of-distribution chemistry. A comprehensive calibration study for neural network structure–activity models reports that model selection strategy and calibration approaches significantly affect reliability of uncertainty estimates. Ensemble-based uncertainty and recalibration frameworks in molecular property prediction show that recalibration can improve uncertainty quality on held-out data, including under distribution shift, but at higher training cost.

5.4 DDI Benchmarks as a Concrete Fusion Setting

DDI prediction is a canonical scenario where representations of two entities (drug A, drug B) and often multiple feature types (structure/fingerprints/-targets/enzymes) are fused. Benchmarking under distribution changes between known and new drug sets emphasises that evaluation must consider shift. Practical fingerprint-only DDI work explicitly tests pairwise fusion operators (addition, subtraction, Hadamard product) for representing drug pairs, directly relevant to unbindable entity fusion of two ECFP vectors.

6 Calibration Diagnostics and Metrics

6.1 Reliability Diagrams

A reliability diagram bins predictions by confidence and plots empirical accuracy vs. predicted confidence. The diagonal indicates perfect calibration; below-diagonal indicates overconfidence; above-diagonal indicates underconfidence.

6.2 Scalar Calibration Metrics

Table 2 summarises the scalar metrics with practical guidance for reporting.

6.3 Recommended Chart Set

For a calibration-focused evaluation, a defensible minimal set is:

1. Reliability diagram (before/after calibration),
2. Bar chart of ECE and MCE across methods,
3. Line or dot chart of Brier and NLL across methods,
4. Stratified ECE (e.g., by scaffold bins, cold-start tiers, or time bins).

7 Temperature Scaling and Alternatives

7.1 Temperature Scaling as the Required Baseline

Temperature scaling is well suited when fusion produces mostly global overconfidence—a common pattern when adding features increases margin without proportionally increasing correctness. Key properties:

- **Low variance / low overfitting risk**: one parameter T, fitted on a held-out calibration set.
- **Argmax-preserving for multiclass**: rankings of logits are unchanged by division by T, so top-1 accuracy typically stays constant.
- **Optimises NLL**: connects cleanly to proper scoring rules.

7.2 Variants and Alternatives

Table 3 contrasts temperature scaling variants and common alternative calibrators.

7.3 Fusion-Aware Calibration Strategies

For fusing two unbindable sources, calibration often improves if you calibrate in stages:

1. **Branch calibration**: calibrate per-modality logits (or per-entity outputs) before fusion.
2. **Fusion calibration**: fuse (e.g., logit averaging or learned weights), then apply final TS.

A practical variant for DDI/pair tasks with two molecules: use permutation-invariant pair representations (e.g., $\mathbf{f}_a + \mathbf{f}_b$, $|\mathbf{f}_a - \mathbf{f}_b|$, $\mathbf{f}_a \odot \mathbf{f}_b$) to reduce spurious ordering effects, then calibrate.

8 Experimental Design Blueprint

This section gives a deployment-ready experimental plan that directly targets calibration degradation from fused ECFP outputs and tests temperature scaling as the first mitigation.

8.1 Candidate Datasets and Splits

Table 4 proposes datasets covering the regimes where calibration often fails.

8.2 Fusion Strategies to Compare

For two-molecule ECFP fusion (DDI-style):

- Concatenation $[\mathbf{f}_a \| \mathbf{f}_b]$ plus order augmentation.
- Permutation-invariant operators: $\mathbf{f}_a + \mathbf{f}_b$, $|\mathbf{f}_a - \mathbf{f}_b|$, $\mathbf{f}_a \odot \mathbf{f}_b$.
- Bilinear/outer-product interactions (higher capacity; likely worse calibration unless regularised).

For heterogeneous fusion (ECFP + learned embedding):

- Early fusion: concat at feature level into an MLP head.
- Late fusion: separate heads produce logits, then combine and apply TS post-hoc.
- Gated fusion: learn per-sample modality weights (high risk for miscalibration if gate becomes overconfident).

8.3 Recommended Experimental Matrix

Table 5 defines the minimal but informative experimental design.

Table 2: Calibration metrics: properties and reporting guidance.

Metric	**What It Captures**	**Strengths**	**Caveats**	**Recommended Reporting**
Reliability diagram	Shape of miscalibration vs confidence	Diagnoses over/underconfidence; communicable	Bin choice; hides sample counts	Plot with bin counts overlay; stratify by subset (OOD/cold-start)
ECE	Average calibration gap	Simple scalar; widely used	Binning bias; can mask local errors	Specify binning (equal-width vs equal-mass), #bins, CI via bootstrap
MCE	Worst-bin gap	Sensitive to failure regions	High variance with small bins	Report with ECE; show where worst bin occurs
Brier	Probabilistic accuracy (squared error)	Proper score; decomposable	Conflates calibration + refinement	Report macro and micro variants; alongside NLL
NLL	Proper score aligned with TS fitting	Directly optimised in TS	Sensitive to label noise	Report pre/post TS; stratify by subset; include ECE diagrams

Table 3: Calibration methods: comparison for fused ECFP pipelines.

Method	**Space**	**Params**	**Argmax?**	**Data Need**	**Best For**
Temperature scaling (TS)	logits	1	Yes	Small	Global over/underconfidence
Vector / classwise scaling	logits	$O(C)$	Usually	Moderate	Class-dependent bias
Matrix scaling	logits	$O(C^2)$	Not guaranteed	Large	Rich miscalibration (risky with few classes)
Local / adaptive TS	logits	>1	Often	Moderate–large	Region-dependent miscalibration
Platt scaling	score→prob	2	N/A	Small–moderate	Binary calibration; margin-based models
Isotonic regression	score→prob	nonparam.	N/A	Large	Complex monotone calibration curves
Dirichlet calibration	probs/logits	$O(C^2)$	Not guaranteed	Moderate–large	Multiclass calibration beyond single T
Deep ensembles	predictive dist.	$K\times$model	N/A	Large compute	Robustness under dataset shift

Table 4: Recommended datasets and split strategies for calibration evaluation.

Dataset	**Task Type**	**Why Useful for Calibration Study**	**Recommended Splits**
DrugBank multi-type DDI	Multiclass (86 types)	Natural two-entity fusion (drug A + drug B) with ECFP; many labels; real shift concerns	Drug-disjoint cold-start; distribution-change splits
MoleculeNet (BBBP, BACE, HIV, ClinTox)	Binary/multitask	Standard chemical benchmarks; supports fingerprint baselines and scaffold splits	Scaffold split; random split for contrast
TDC (ADMET, toxicity)	Classification/regression	Curated tasks with explicit emphasis on distribution shifts	Scaffold / temporal / TDC split functions
ChEMBL target-wise activity	Binary per target	Large-scale structure–activity data; suited to time-based drift	Time split by assay; scaffold split

Table 5: Experimental matrix for calibration evaluation.

Component	**Baseline**	**Fusion Variants**	**Calibration**	**What You Learn**
Representation	ECFP only	ECFP(A)+ECFP(B) or ECFP + embedding	None	Whether fusion improves discrimination
Pair fusion operator	Concat	Sum / abs-diff / Hadamard	TS	Whether permutation invariance improves calibration–discrimination trade-off
Heterogeneous fusion	Early concat	Late logit fusion; gated fusion	TS; adaptive/local TS	Whether miscalibration is global vs region-dependent
Calibration complexity	TS	—	Vector scaling; Dirichlet; isotonic	When single temperature is insufficient
Shift robustness	i.i.d.	Scaffold/cold-start/time	TS vs ensembles	Whether post-hoc calibration survives shift

9 Implementation Details

9.1 ECFP Generation Parameters

Recommended ranges to justify empirically:

- Radius: 2 (ECFP4) or 3 (ECFP6).
- fpSize: 1024 / 2048 / 4096.
- Include chirality: on/off depending on task.
- Bit vs count: both worth testing if calibration is sensitive to sparsity.

9.2 Temperature Scaling Optimisation

Practical guidelines:

- Optimise $\log T$ to enforce $T > 0$.
- Initial $T = 1$.
- Optimiser: L-BFGS (common) or Adam.
- Clamp or regularise T to a sane range such as $[0.05, 20]$ to avoid numerical pathologies.
- Use float64 during calibration fitting if logits are large.

9.3 Pseudocode: Multiclass Temperature Scaling

```
Input:
  trained_model f(x) -> logits z in R^C
  calibration dataset D_cal = {(x_i, y_i)}
Goal:
  fit scalar T > 0 to minimise NLL on D_cal
Procedure:
  1) Freeze f
  2) For each x_i in D_cal:
       z_i = f(x_i) # logits
  3) Optimise logT:
       T = exp(logT)
       p_i = softmax(z_i / T)
       loss = mean_i( -log p_i[y_i] )
  4) Return calibrated predictor:
       p_T(x) = softmax(f(x) / T)
```

Listing 1: Temperature scaling procedure.

9.4 PyTorch Implementation

```python
import torch
import torch.nn.functional as F

class TemperatureScaler(torch.nn.Module):
    def __init__(self, init_temp=1.0):
        super().__init__()
        self.log_T = torch.nn.Parameter(
            torch.tensor([init_temp]).log())

    @property
    def T(self):
        return self.log_T.exp()

    def forward(self, logits):
        return logits / self.T
```

```python
def fit_temperature(model, calib_loader,
                    device="cuda"):
    model.eval()
    logits_list, y_list = [], []
    with torch.no_grad():
        for x, y in calib_loader:
            x = x.to(device)
            y = y.to(device)
            logits_list.append(model(x))
            y_list.append(y)
    logits = torch.cat(logits_list)
    y = torch.cat(y_list)

    scaler = TemperatureScaler(1.0).to(device)
    opt = torch.optim.LBFGS(
        scaler.parameters(),
        lr=0.1, max_iter=50)

    def closure():
        opt.zero_grad()
        scaled = scaler(logits)
        loss = F.cross_entropy(scaled, y)
        loss.backward()
        return loss

    opt.step(closure)
    return scaler
```

Listing 2: Minimal temperature scaling in PyTorch.

9.5 Multilabel Note

If the fused ECFP model is multilabel (e.g., multiple DDI types can co-occur), independent sigmoid outputs are typical. In that case a single global T on logits may be too coarse; consider per-label temperature or Platt scaling, but guard against overfitting for rare labels.

10 Expected Results and Interpretation

10.1 What to Expect

In many fused-ECFP settings:

- **Discrimination improves, calibration worsens** after fusion, because the fused model becomes more confident faster than it becomes more correct. This is consistent with findings relating miscalibration to model capacity and hard-label training.
- **Temperature scaling reduces ECE/MCE and improves NLL**, often substantially, with negligible effect on accuracy for multiclass tasks. If fitted $T > 1$, the model was overconfident; $T < 1$ indicates underconfidence.
- Under **distribution shift**, improvements from TS may shrink or disappear; this supports in-

cluding ensemble baselines or shift-aware evaluation.

10.2 Limitations and Open Questions

- **ECE is not a gold standard**: its estimate depends on binning and sample size; complementing ECE with proper scoring rules and reliability plots is important.
- **Single-temperature models cannot fix structural miscalibration** (classwise, subsetwise, or confidence-conditional). If fusion introduces heterogeneous error modes, more expressive calibrators may be required.
- **Calibration under long-tail labels** is particularly hard: calibration can look good overall while failing catastrophically on rare classes; classwise ECE and stratified analyses are necessary.
- **Fusion-specific desiderata** (e.g., confidence should not increase when a modality is removed) are not captured by standard ECE/Brier alone.

11 Recommendations for Evaluation Writing

A strong evaluation narrative follows this structure:

1. Start with a crisp statement: "Fusion improves discrimination but can degrade calibration; for deployment-like decisions we need calibrated probabilities." Ground it with core calibration references and at least one domain-specific reference.
2. Define the fusion setting precisely: pairwise fusion (two drugs) vs. heterogeneous fusion (ECFP + embedding), and state why the sources are unbindable.
3. Present a short theory-of-failure table (Table 1), then the experimental matrix.
4. Report calibration with **both plots and proper scores** (reliability diagrams + ECE/MCE + Brier + NLL), and be explicit about binning and calibration-set design.
5. Treat temperature scaling as the *required baseline*; then motivate any more complex calibrator only if TS fails in identifiable ways (e.g., classwise miscalibration, shift-specific failures).

11.1 Reproducibility Checklist

To make the evaluation credible, specify:

- Exact split definitions (random vs scaffold vs cold-start vs time/distribution-change).
- Calibration set size and sampling scheme (especially in long-tail label regimes).
- Bin count and binning strategy for ECE/MCE; bootstrap CIs.
- Whether calibration is fit once per run, per fold, or per seed (recommended: per fold).

11.2 Connection to the Cold-Drug DDI Benchmark

In the context of the cold-drug DDI benchmark presented in the companion paper, the calibration findings are concrete: adding ECFP4 fusion to the GAT encoder improves macro PR-AUC by +6.1 pp and accuracy by +3.8 pp, but ECE degrades from 0.1394 to 0.1966 (+41% relative). This pattern matches the "increased capacity via fusion" cause (Section 3): the fused model learns sharper decision boundaries from deterministic ECFP features, producing overconfident softmax outputs. Temperature scaling on the validation calibration split is the recommended first step; if classwise analysis reveals that tail classes (bottom 18 by training support) are disproportionately miscalibrated, vector scaling or classwise temperature should be evaluated.